Enterprise Data Quality Assessment & Remediation for NYC 311 Service Requests

A SQL Driven Data Cleaning Project with Responsible AI Augmentation

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Executive Summary

Data quality issues can significantly impact operational reporting, analytical accuracy, and decision making. This project implemented a structured Data Quality Assessment and Remediation Framework on the NYC 311 Service Requests dataset using SQL Server and a rule based data quality methodology.

The project assessed over **5.3 million service request records**, identified quality issues through systematic profiling, developed a formal Data Quality Rulebook, executed remediation activities, and validated improvements through post clean profiling and rule based testing.

The framework successfully improved data quality across multiple dimensions, including uniqueness, validity, consistency, and timeliness, while maintaining transparency through exception management and auditability.

Key outcomes include:

- 100,984 duplicate records identified and removed
- 26,711 status inconsistencies resolved
- 1,097 invalid date sequence records removed
- 706 future dated records removed
- 1,040 geographic coordinate integrity violations resolved
- 5 invalid ZIP code records removed
- 100% duplicate elimination
- 100% rule compliance for implemented validation rules

The project demonstrates a repeatable methodology for assessing and improving operational data quality prior to analytical consumption.

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1. Business Problem

Government service request systems collect large volumes of citizen complaint and service response data from multiple channels and agencies.

Over time, operational datasets may accumulate quality issues such as:

- Duplicate records
- Invalid timestamps
- Status inconsistencies

- Geographic inaccuracies
- Inconsistent categorical values
- Invalid reference data

These issues reduce confidence in reporting and can negatively impact downstream analytics.

The objective of this project was to design and implement a structured framework for assessing, quantifying, remediating, and validating data quality issues within the NYC 311 Service Requests dataset.

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2. Project Objectives

The primary objectives were:

1. Assess overall data quality using a structured profiling methodology.
2. Identify data quality issues across multiple dimensions.
3. Develop a formal Data Quality Rulebook.
4. Implement rule-based remediation processes.
5. Validate improvements through post clean profiling.
6. Quantify the impact of cleaning activities.
7. Demonstrate responsible AI augmentation within a modern data workflow.

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3. Dataset Overview

Metric	Value
Dataset	NYC 311 Service Requests
Source	NYC Open Data
Date Range	2025-01-01 to 2026-06-01
Raw Records	5,314,955
Clean Records	5,206,784
Records Removed / Quarantined	108,171
Initial Columns	12
Final Columns	11

The dataset contains citizen service requests submitted to various New York City agencies and includes complaint classifications, geographic information, service status information, and lifecycle timestamps.

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4. Methodology

A structured data quality lifecycle was implemented.

Raw Ingestion
↓
Data Profiling
↓
Data Quality Assessment
↓
Rulebook Development
↓
Data Cleaning
↓
Exception Management
↓
Validation
↓
Quality Scorecard
↓
Reporting

The methodology follows principles commonly used within enterprise data governance and data quality programs.

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5. Data Profiling Approach

Data profiling was performed prior to any transformation activities.

The profiling process assessed:

Completeness

- NULL value analysis
- Missing value distribution

Uniqueness

- Duplicate record detection
- Duplicate rate measurement

Validity

- Date validation
- ZIP code validation
- Geographic validation

Consistency

- City standardization assessment
- Borough standardization assessment
- Complaint type standardization assessment

Timeliness

- Future date validation
- Service request lifecycle validation

Profiling findings formed the basis of all remediation activities.

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6. Data Quality Rulebook

A formal Data Quality Rulebook containing 20 validation rules was developed.

Rules were categorized according to severity:

Critical

Violations impacting:

- Data integrity
- Record identification
- Business process reporting

High

Violations impacting:

- Analytics
- Geographic reporting
- Operational decision-making

Medium

Violations impacting:

- Standardization
- Consistency

Low

Violations impacting:

- Presentation quality

Examples include:

- Unique Service Request Identifier

- Closed Date Validation
- Status Consistency
- Geographic Coordinate Completeness
- ZIP Code Validation
- City Standardization
- Borough Standardization
- Complaint Type Standardization

The rulebook served as the foundation for remediation and validation activities.

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7. Data Cleaning Methodology

The cleaning process was designed using findings from the profiling phase and corresponding data quality rules.

Duplicate Removal

Removed duplicate records while excluding the unique_key field and retaining the most recent record.

Date Validation

Removed records where:

- created_date > closed_date
- created_date > due_date

Future Date Validation

Removed records where:

- closed_date > '2026-06-03'

Status Reconciliation

Updated status values to Closed where:

- closed_date IS NOT NULL
- status <> 'Closed'

Closed Status Validation

Removed records where:

- status = 'Closed'
- closed_date IS NULL

Standardization

Implemented mapping tables for:

- City
- Borough
- Complaint Type

ZIP Code Validation

Removed invalid ZIP codes while preserving NULL values.

Data Type Enforcement

Converted ingestion-layer VARCHAR columns into appropriate analytical datatypes.

Post Clean Validation

Re ran duplicate checks and validation rules following all transformations.

Column Rationalization

Removed due_date from the analytical layer due to insufficient population and limited analytical value.

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8. Exception Management

Records removed during remediation were not discarded without traceability.

An exception table was created containing:

- unique_key
- exception reason

This approach preserves lineage and enables all removed records to be traced back to the Bronze layer for auditing and review.

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9. Cleaning Impact Analysis

Cleaning Activity	Records Affected
Duplicate Removal	100,984
Invalid Date Sequence Removal	1,097
Future Date Removal	706
Status Corrections	20,672
Closed Status Violations Removed	6,039
Coordinate Integrity Violations Removed	1,040
Invalid ZIP Codes Removed	5

Total records removed or quarantined: 108,171

10. Rule Validation Results

Following remediation, all implemented validation rules were re-executed against the cleaned dataset.

Key results include:

Duplicate Validation

Before:

- 100,984 duplicate records

After:

- 0 duplicate records

Date Validation

Before:

- 1,097 invalid date sequences

After:

- 0 violations

Status Validation

Before:

- 26,711 status inconsistencies

After:

- 0 violations

Coordinate Validation

Before:

- 1,040 coordinate integrity violations

After:

- 0 violations

ZIP Code Validation

Before:

- 5 invalid ZIP codes

After:

- 0 invalid ZIP codes

All implemented validation rules successfully passed in the cleaned dataset.

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11. Data Quality Scorecard

Dimension	Raw Dataset	Clean Dataset
Completeness	98.86%	98.88%
Uniqueness	98.10%	100.00%
Validity	99.83%	100.00%
Timeliness	99.99%	100.00%

Key observations:

- Completeness remained stable because missing values were preserved rather than imputed.
- Uniqueness improved through duplicate elimination.
- Validity improved through rule based remediation.
- Timeliness improved through future date validation.

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12. Responsible AI Augmentation

Artificial Intelligence was incorporated throughout the project as a productivity enhancement tool.

Project Planning

- Generated project workplan
- Generated data quality workflow

Architecture and Ingestion

- Generated Bronze, Silver, and Gold schemas
- Generated raw table creation scripts
- Generated bulk insert scripts
- Assisted with bulk insert troubleshooting

Profiling and Validation

- Generated profiling workflow
- Generated NULL analysis scripts
- Assisted with rule development

Standardization

- Consolidated city variations
- Consolidated borough variations

- Consolidated complaint type variations

SQL Development

- Generated mapping logic
- Generated datatype conversion scripts
- Generated validation reporting templates

Reporting

- Generated rule validation framework
- Generated cleaning impact analysis framework

Human Responsibilities

The analyst remained responsible for:

- Dataset selection
- Profiling execution
- Interpretation of findings
- Remediation decisions
- Validation
- Quality assessment
- Final conclusions

No operational records were submitted for automated analysis. AI was used exclusively for workflow design, SQL generation, troubleshooting, and documentation support.

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13. Key Learnings

Data Profiling Drives Effective Cleaning

The most effective cleaning actions emerged directly from profiling findings rather than assumptions.

Rule Based Validation Improves Auditability

A formal rulebook enabled repeatable validation and transparent remediation.

Exception Management Preserves Lineage

Logging exceptions rather than silently deleting records improved traceability and governance.

Standardization Improves Analytical Consistency

Reference mappings significantly reduced categorical inconsistencies while preserving business meaning.

AI Accelerates Development

AI substantially reduced development effort while still requiring human validation and oversight.

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14. Future Improvements

Future iterations could include:

- Automated rule execution pipelines
- Data quality monitoring dashboards
- Data observability metrics
- Geographic reference data validation
- Incremental quality monitoring
- Enterprise governance integration
- Automated exception reporting

Conclusion

This project successfully implemented a comprehensive Data Quality Assessment and Remediation Framework on over 5.3 million NYC 311 service request records.

Through structured profiling, rulebook development, remediation, validation, and exception management, the framework eliminated duplicate records, resolved status inconsistencies, corrected date violations, improved categorical consistency, and enhanced overall data quality.

The resulting methodology provides a repeatable and auditable approach to preparing operational datasets for analytical consumption while demonstrating practical SQL development, data quality management, governance awareness, and responsible AI augmentation.